Visvesvaraya Technological University, Belagavi

REPORT

On

On-line Monitoring System for Photovoltaic Power Generation Power Quality Based onESP32

For the academic year 2025-26

Submitted by

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CERTIFICATE

This is to certify that the Project Work entitled **“On-line Monitoring System for Photovoltaic Power Generation Power Quality Based on ESP32”** carried out by **Prathika R P**, **Prathiksha**, and **Priyanka R** bearing USN **1CR24EC155**, **1CR24EC156**, and **1CR24EC159** respectively, are bonafide students of CMR Institute of Technology, Bengaluru, in partial fulfillment for the award of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi, during the academic year **2025–26**.

It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the departmental library. The report has been approved as it satisfies the academic requirements in respect of the work prescribed for the said degree.

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ABSTRACT

The growing demand for renewable energy sources has led to the widespread adoption of photovoltaic (PV) power generation systems. Solar energy is one of the most promising and environmentally friendly energy sources available today. However, maintaining the power quality and operational efficiency of photovoltaic systems is a major challenge due to variations in environmental conditions, load fluctuations, and electrical disturbances. Continuous monitoring of electrical parameters is essential to ensure reliable operation, efficient energy utilization, and early detection of faults in PV power generation systems.

This project presents the design and development of an **On-line Monitoring System for Photovoltaic Power Generation Power Quality Based on ESP32**. The proposed system continuously monitors important electrical parameters of a photovoltaic power generation system such as voltage, current, power, frequency, and other power quality indicators. The ESP32 microcontroller serves as the main processing unit and acquires data from the connected sensing modules. The measured values are processed and analyzed in real time to evaluate the operating condition and performance of the PV system.

The monitored parameters are displayed to the user and can be transmitted through wireless communication features available in the ESP32. This enables remote observation of the system without the need for direct physical access to the installation site. Real-time monitoring helps identify abnormal operating conditions, voltage fluctuations, power losses, and other factors that may affect system efficiency and reliability.

The proposed system is designed to be cost-effective, easy to implement, and suitable for small-scale as well as medium-scale photovoltaic installations. By continuously monitoring power quality parameters, the system assists users in maintaining optimal system performance and reducing maintenance costs.

Several research studies on photovoltaic power generation systems emphasize the importance of continuous monitoring and power quality assessment. Poor power quality can lead to reduced efficiency, equipment malfunction, increased energy losses, and shortened system life. Existing monitoring solutions often involve expensive equipment and complex installation procedures. Recent advancements in embedded systems and wireless communication technologies have enabled the development of low-cost monitoring systems using microcontrollers such as ESP32.

Literature studies indicate that real-time monitoring of photovoltaic systems improves system reliability, enhances energy management, and enables early fault detection. Researchers have

demonstrated that monitoring electrical parameters such as voltage, current, power output, and frequency provides valuable information regarding the overall health and performance of solar power generation systems. IoT-enabled monitoring platforms further improve accessibility by allowing users to view system information remotely.

The developed online monitoring system effectively measures and analyzes key power quality parameters of photovoltaic power generation systems. Experimental observations show that the ESP32-based monitoring system provides reliable and continuous measurement of electrical parameters under different operating conditions. The system successfully identifies variations in voltage, current, and power output caused by changing environmental conditions and load requirements.

The proposed solution offers an efficient approach for monitoring photovoltaic power generation systems and maintaining power quality standards. It improves operational reliability, supports preventive maintenance, reduces downtime, and contributes to the effective utilization of renewable energy resources. The system can be further enhanced through cloud connectivity, advanced data analytics, mobile application support, and intelligent fault prediction techniques, making it suitable for future smart energy management applications.

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CHAPTER 1

INTRODUCTION

Photovoltaic (PV) power generation is one of the most widely used renewable energy technologies for producing clean and sustainable electricity. The performance of PV systems depends on various factors such as solar irradiance, temperature, and load conditions. Continuous monitoring of electrical parameters is necessary to ensure efficient operation and maintain power quality.

The proposed project, **“On-line Monitoring System for Photovoltaic Power Generation Power Quality Based on ESP32”**, is designed to monitor important electrical parameters such as voltage and current in real time. The ESP32 microcontroller collects sensor data, processes it, and displays the measured values for analysis and monitoring.

1.1 MOTIVATION

The increasing use of solar energy systems has created a need for continuous monitoring of their performance. Variations in environmental conditions and electrical faults can reduce system efficiency. This project is motivated by the need to develop a simple and cost-effective system that continuously monitors photovoltaic power generation parameters and helps improve system reliability.

1.2 PROBLEM STATEMENT

Photovoltaic systems often operate under changing environmental conditions, which can affect their performance and power quality. Traditional monitoring methods are expensive and require manual inspection. Therefore, there is a need for a low-cost monitoring system that can continuously measure and display important electrical parameters in real time.

1.3 OBJECTIVE OF PROJECT

• To design an online monitoring system for photovoltaic power generation using ESP32.

• To measure voltage and current parameters of the photovoltaic system.

• To monitor power generation performance in real time.

• To display electrical parameters continuously for analysis.

• To improve system reliability through continuous monitoring.

• To identify abnormal operating conditions and potential faults.

• To provide a low-cost and efficient monitoring solution for photovoltaic systems.

1.4 LITERATURE SURVEY

Several research studies have focused on monitoring and improving the performance of photovoltaic power generation systems. Researchers have demonstrated that continuous monitoring of voltage, current, power output, and environmental conditions significantly improves system efficiency and reliability.

Recent developments in IoT-based monitoring systems have enabled real-time observation of photovoltaic installations using wireless communication technologies. ESP32-based monitoring systems have gained popularity because of their low cost, compact size, and built-in communication features. Voltage and current sensors are commonly used to collect electrical parameters, which are then processed and displayed for performance evaluation.

The literature indicates that real-time monitoring plays an important role in fault detection, preventive maintenance, and optimization of photovoltaic power generation systems. These studies highlight the need for simple, cost-effective, and reliable monitoring solutions that can be implemented in both small-scale and large-scale solar installations.

1.5 DESIGN APPROACH

Hardware Design and Development

1. Voltage Measurement: The voltage sensor continuously measures the output voltage of the photovoltaic system.

2. Current Measurement: The ACS712 current sensor measures the output current flowing through the system.

3. Data Acquisition: The ESP32 collects sensor readings through its analog input pins.

4. Data Processing: The acquired data is processed to calculate electrical parameters and evaluate system performance.

5. Display Unit: The measured values are displayed on an LCD module for easy observation.
6. Communication: The ESP32 can transmit monitored data through its wireless communication capability for remote monitoring.

Testing

• Sensor Calibration: Voltage and current sensors are calibrated to ensure accurate measurements.

• Functional Testing: Individual modules are tested separately before integration.

• Real-Time Testing: The complete system is tested under different operating conditions.

• Accuracy Testing: Measured values are compared with standard measuring instruments.

System Testing

The complete system is tested to verify accurate voltage and current measurement, proper display operation, and reliable monitoring of photovoltaic power generation parameters.

2.1 BLOCK DIAGRAM

The block diagram shown in Fig. 2.1 represents the **On-line Monitoring System for Photovoltaic Power Generation Power Quality Based on ESP32**. The ESP32 acts as the main controller and performs all monitoring and processing operations of the system.

A ZMPT101B voltage sensor is connected to the ESP32 to measure the AC voltage of the system. The ACS712 current sensor is used to measure the load current flowing through the circuit. The measured voltage and current values are continuously sent to the ESP32 for processing.

The ESP32 calculates important electrical parameters such as voltage, current, power, and energy consumption. A 16×2 LCD display is interfaced with the ESP32 to display the measured values in real time.

A relay module is connected to the system for switching and control operations. The inverter converts the DC power obtained from the battery or photovoltaic source into AC power required for monitoring and testing purposes.

The sensors, LCD display, relay module, and inverter receive the necessary power supply for proper operation. The ESP32 continuously monitors the sensor values, processes the collected data, and displays the output on the LCD screen.

Thus, the proposed system enables real-time monitoring of photovoltaic power generation parameters and helps in analyzing the power quality of the system.

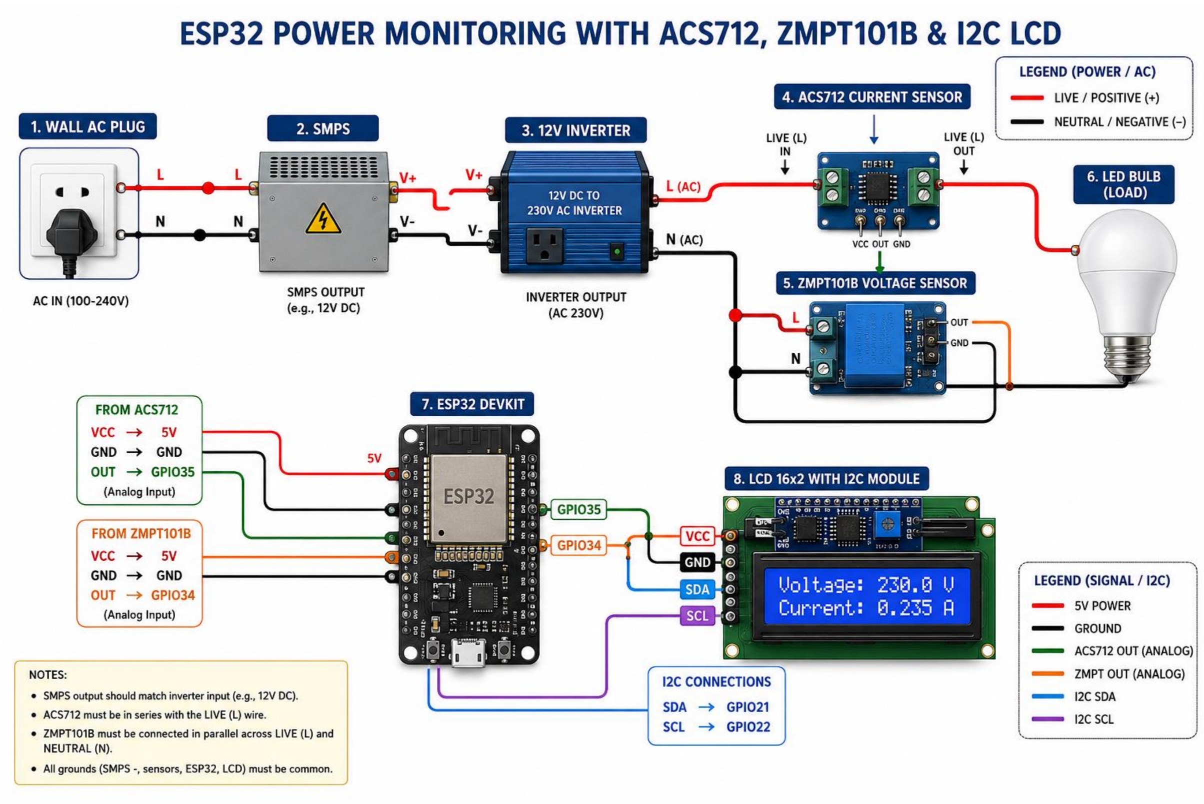

Figure 2.1: Block Diagram

2.2 ESP32 MICROCONTROLLER

The ESP32 is a low-cost and high-performance microcontroller developed by Espressif Systems.

It is widely used in embedded systems, Internet of Things (IoT) applications, and real-time monitoring systems due to its powerful processing capability and built-in Wi-Fi and Bluetooth features.

In the proposed system, the ESP32 acts as the main controller. It receives voltage and current data from the ZMPT101B and ACS712 sensors, processes the acquired data, calculates electrical parameters such as voltage, current, power, and energy, and displays the results on the LCD display.

The ESP32 offers multiple analog and digital input/output pins, making it suitable for interfacing with various sensors and peripheral devices. Its low power consumption, compact size, and wireless communication capabilities make it an ideal choice for photovoltaic power monitoring applications.

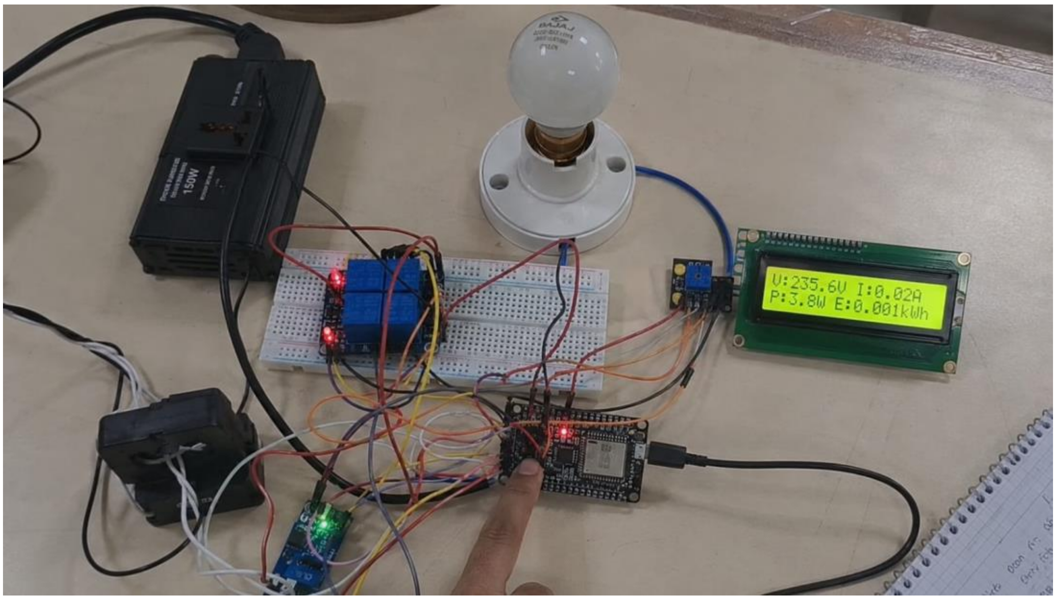

Figure 2.2: ESP32 Development Board

2.2.1 ESP32 DETAILS

The ESP32 is based on a dual-core processor and operates at a voltage of 3.3V. It supports Wi-Fi and Bluetooth communication and provides multiple GPIO pins for interfacing external devices.

Specifications:

• Operating Voltage : 3.3V

• Input Voltage : 5V

• Dual-Core Processor

• Built-in Wi-Fi and Bluetooth

• Multiple GPIO Pins

• Low Power Consumption

2.2.2 FEATURES OF ESP32

• High processing speed

• Built-in Wi-Fi and Bluetooth connectivity

• Supports analog and digital interfacing

• Low power consumption

• Compact size and low cost

• Suitable for IoT and monitoring applications

2.2.3 ESP32 PIN DESCRIPTION

The ESP32 contains several analog and digital input/output pins. Analog pins are used to acquire data from the voltage and current sensors, while digital pins are used for LCD communication and relay control. The GPIO pins provide flexible interfacing options for different sensors and modules used in the project.

2.3 ZMPT101B VOLTAGE SENSOR

The ZMPT101B is a precision AC voltage sensor module used for measuring AC voltage in power monitoring applications. It provides an analog output proportional to the input voltage and offers high measurement accuracy. In the proposed system, the ZMPT101B sensor is used to measure the AC voltage generated by the inverter.

2.3.1 ZMPT101B SENSOR DETAILS

• Operating Voltage : 5V

• High Measurement Accuracy

• Adjustable Sensitivity

• Analog Output

• Suitable for AC Voltage Measurement

2.3.2 WORKING PRINCIPLE OF ZMPT101B SENSOR

The ZMPT101B sensor contains a precision voltage transformer that senses the AC voltage. The sensor converts the measured voltage into an analog signal which can be processed by the ESP32 microcontroller.

2.3.3 ZMPT101B INTERFACING WITH ESP32

The sensor output pin is connected to one of the analog input pins of the ESP32. The ESP32 continuously reads the sensor output and calculates the corresponding voltage value.

2.4 ACS712 CURRENT SENSOR

The ACS712 is a Hall-effect based current sensor used for measuring both AC and DC current. It is widely used in energy monitoring systems because of its reliability and ease of interfacing.

2.4.1 ACS712 SENSOR DETAILS

- Operating Voltage : 5V
- Measures AC and DC Current
- Hall-Effect Based Sensor
- Analog Output
- High Accuracy

2.4.2 WORKING PRINCIPLE OF ACS712 SENSOR

The ACS712 sensor detects the magnetic field produced by the current flowing through a conductor. This magnetic field is converted into a proportional analog voltage which is measured by the ESP32.

2.4.3 ACS712 INTERFACING WITH ESP32

The output pin of the ACS712 sensor is connected to an analog input pin of the ESP32. The ESP32 processes the sensor output and calculates the current flowing through the load.

2.5 16×2 LCD DISPLAY

The 16×2 LCD display is used to display voltage, current, power, and energy values measured by the system. It provides a simple and effective user interface for real-time monitoring.

2.5.1 LCD DISPLAY DETAILS

- 16 Characters × 2 Lines
- Operating Voltage : 5V
- Low Power Consumption
- Easy Interfacing
- Real-Time Display

2.5.2 LCD DISPLAY INTERFACING WITH ESP32

The LCD display is connected to the ESP32 through the I2C communication interface. The ESP32 sends processed sensor data to the LCD, which displays the measured electrical parameters.

2.6 RELAY MODULE

A relay module is an electrically operated switch used for controlling electrical loads. In this project, the relay module is used for switching and protection purposes.

2.6.1 RELAY MODULE DETAILS

• Operating Voltage : 5V

• Electromagnetic Switching Device

• Provides Electrical Isolation

• Suitable for AC and DC Loads

2.6.2 WORKING PRINCIPLE OF RELAY MODULE

When a control signal is applied from the ESP32, the relay coil gets energized and changes the switching position of the relay contacts. This allows the connected load to be controlled safely.

2.6.3 RELAY MODULE INTERFACING WITH ESP32

The control input of the relay module is connected to a digital output pin of the ESP32. The relay operates according to the control signal generated by the microcontroller.

2.7 INVERTER

An inverter is an electrical device that converts DC power into AC power. In the proposed system, the inverter converts the DC supply from the battery into AC voltage for monitoring and testing purposes.

2.7.1 INVERTER DETAILS

• Input : DC Voltage

• Output : AC Voltage

• Compact Design

• High Efficiency

2.7.2 ROLE OF INVERTER IN THE PROJECT

The inverter provides the AC voltage required for monitoring. The voltage and current sensors measure the inverter output, enabling analysis of electrical parameters and power quality.

2.8 JUMPER WIRES

Jumper wires are electrical connecting wires used to establish temporary connections between various electronic components. They are widely used during circuit assembly and testing.

2.8.1 JUMPER WIRES DETAILS

• Flexible and Reusable

• Available in Different Colours

• No Soldering Required

• Easy Circuit Modification

2.8.2 JUMPER WIRES INTERFACING IN ESP32 PROJECTS

Jumper wires are used to connect the ESP32, sensors, LCD display, relay module, and power supply. Different coloured wires are generally used for power, ground, and signal connections.

CHAPTER 3

SOFTWARE DESIGN

3.1 HARDWARE RESOURCE ALLOCATION

ESP32 Pin	Component	Description
GPIO34	ZMPT101B	Voltage Sensor Input
GPIO35	ACS712	Current Sensor Input
GPIO21	SDA	LCD Communication
GPIO22	SCL	LCD Communication
GPIO23	Relay Module	Relay Control

3.2 PROGRAMMING LANGUAGE

The proposed system is programmed using Embedded C/C++ language in Arduino IDE. The language provides simple syntax and efficient hardware control for embedded applications.

3.3 DEVELOPMENT TOOLS

• Arduino IDE

• ESP32 Board Package

• Serial Monitor

• LCD Library

3.4 ARDUINO IDE

Arduino IDE is an open-source software used to write, compile, and upload programs to ESP32. It provides a user-friendly environment for embedded system development.

3.5 EMBEDDED C PROGRAMMING

Embedded C is used to acquire sensor data, process electrical parameters, control the LCD display, and manage relay operations. The program continuously monitors voltage, current, power, and energy values.

CHAPTER 4

APPLICATIONS

4.1 APPLICATIONS

• Solar Power Monitoring Systems

• Renewable Energy Projects

• Educational Laboratories

• Energy Management Systems

• Industrial Power Monitoring

4.2 ADVANTAGES

• Real-time Monitoring

• Low Cost

• High Reliability

• Easy Implementation

• Compact Design

• Accurate Measurement

4.3 DISADVANTAGES

• Sensor Calibration Required

• Accuracy Depends on Sensor Quality

• Limited to Small-Scale Monitoring Applications

• External Power Supply Required

CHAPTER 5

RESULTS AND TESTING

5.1 MODULE TESTING

Module testing is performed to verify the operation of individual hardware components before integrating the complete system.

5.1.1 POWER SUPPLY TESTING

The power supply is tested using a multimeter to verify the required voltage levels. A stable power supply ensures proper operation of the ESP32, sensors, LCD display, and relay module.

5.1.2 ZMPT101B VOLTAGE SENSOR TESTING

The ZMPT101B voltage sensor is connected to the ESP32 and tested by measuring the AC voltage from the inverter output. The sensor readings are observed and verified for proper operation.

5.1.3 ACS712 CURRENT SENSOR TESTING

The ACS712 current sensor is tested by connecting different electrical loads. The sensor successfully measures the load current and provides corresponding analog output to the ESP32.

5.1.4 LCD DISPLAY TESTING

The LCD display is tested by displaying sample messages and measured values. Proper display of voltage, current, power, and energy confirms successful communication with the ESP32.

5.1.5 ESP32 BOARD TESTING

The ESP32 board is tested by uploading a simple program and verifying sensor readings. Successful operation confirms the proper functioning of the microcontroller.

5.2 SYSTEM TESTING

5.2.1 ESP32 WITH VOLTAGE SENSOR

The voltage sensor is interfaced with ESP32 and tested under different voltage conditions. The measured values are displayed on the LCD display.

5.2.2 ESP32 WITH CURRENT SENSOR

The current sensor is interfaced with ESP32 and tested with different load conditions. The current values are monitored continuously.

5.2.3 COMPLETE SYSTEM TESTING

The complete system consisting of ESP32, voltage sensor, current sensor, LCD display, relay module, inverter, and load is tested successfully. The system continuously monitors voltage, current, power, and energy values and displays them in real time.

RESULT

The developed system successfully monitors photovoltaic power generation parameters such as voltage, current, power, and energy consumption. The measured values are displayed accurately on the LCD display, demonstrating reliable system performance.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

CONCLUSION

The proposed **On-line Monitoring System for Photovoltaic Power Generation Power Quality Based on ESP32** successfully monitors important electrical parameters such as voltage, current, power, and energy consumption in real time. The ESP32 microcontroller effectively acquires sensor data from the ZMPT101B voltage sensor and ACS712 current sensor, processes the information, and displays the measured values on the LCD display.

The developed system is simple, reliable, cost-effective, and suitable for small-scale photovoltaic monitoring applications. It helps users understand system performance and provides a practical approach for monitoring renewable energy systems.

FUTURE SCOPE

• Integration of IoT technology for remote monitoring.

• Mobile application support for real-time observation.

• Cloud-based data storage and analysis.

• Automatic fault detection and alert generation.

• Improved sensor accuracy and advanced data analytics.

• Monitoring of large-scale photovoltaic installations.

• Wireless monitoring through Wi-Fi and Bluetooth networks.

CHAPTER 7

REFERENCES

1. ESP32 Technical Reference Manual – Espressif Systems.
2. ACS712 Current Sensor Datasheet – Allegro Microsystems.
3. ZMPT101B Voltage Sensor Datasheet.
4. Arduino IDE Documentation.
5. IEEE Research Papers on Photovoltaic Power Monitoring Systems.
6. Renewable Energy Monitoring System Research Articles.
7. Embedded Systems and IoT-Based Energy Monitoring Journals.
8. Electronics Hub and Circuit Digest Technical Articles.

APPENDIX A

COMPONENT LIST

Sl. No	Component Name	Quantity
1	ESP32 Development Board	1
2	ZMPT101B Voltage Sensor	1
3	ACS712 Current Sensor	1
4	16×2 LCD Display	1
5	Relay Module	1
6	Inverter	1
7	Bulb Load	1
8	Breadboard	1

Sl. No	Component Name	Quantity
9	Jumper Wires	As Required
10	Battery / Power Supply	1

APPENDIX B

CODE

The complete Arduino IDE program developed for ESP32 is used to acquire voltage and current sensor readings, calculate power and energy values, control the relay module, and display the measured parameters on the LCD display. The source code is attached in Appendix B for reference and implementation.

APPENDIX B
CODE

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);

const int voltagePin = 34;
const int currentPin = 35;

float voltage = 0;
float current = 0;
float power = 0;
float energy = 0;

unsigned long previousTime = 0;

void setup()
{
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();

  lcd.setCursor(0,0);
  lcd.print("PV Monitoring");
  delay(2000);
  lcd.clear();

  previousTime = millis();
}

void loop()
{
  int voltageRaw = analogRead(voltagePin);
  int currentRaw = analogRead(currentPin);

  voltage = (voltageRaw * 250.0) / 4095.0;
  current = abs((currentRaw - 2048) * 20.0 / 2048.0);

  power = voltage * current;

  unsigned long currentTime = millis();
  float hours = (currentTime - previousTime) / 3600000.0;

  energy += power * hours;

  previousTime = currentTime;

  lcd.setCursor(0,0);
  lcd.print("V:");
  lcd.print(voltage,1);
  lcd.print(" I:");
```

```
lcd.print(current,2);

lcd.setCursor(0,1);
lcd.print("P:");
lcd.print(power,1);
lcd.print("W");

Serial.print("Voltage: ");
Serial.print(voltage);
Serial.print(" V  Current: ");
Serial.print(current);
Serial.print(" A  Power: ");
Serial.print(power);
Serial.println(" W");

delay(1000);
```

}